MILITARY SPECIFICATION

VALVES, BOILER BLOW SERVICE, MONEL, SHIPBOARD USE

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 1-1/2 inch monel (nickel-copper) valves for use in boiler blow systems other than the first valve off of boiler drums and headers, 600 and 1500 pounds per square inch (lb/in^2) service.

1.2 Classification. Monel valves shall be of the straightway and angle design and shall be of the following types, as specified (see 6.2.1).

Type I - Flanged globe, T pattern, 600 lb/in^2 service
Type II - Flanged angle, 600 lb/in^2 service
Type III - Flanged globe stop check, T pattern, 600 lb/in^2 service
Type IV - Flanged angle stop check, 600 lb/in^2 service
Type V - Flanged globe, T pattern, 1500 lb/in^2 service
Type VI - Flanged angle, 1500 lb/in^2 service
Type VII - Flanged globe stop check, T pattern, 1500 lb/in^2 service
Type VIII - Flanged angle stop check, 1500 lb/in^2 service

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-C-523	- Copper Alloy Ingots; Brass (Yellow, High Strength) Manganese and Manganese-Aluminum-Bronze (for Remelting).
QQ-N-281	- Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-286	- Nickel-Copper-Aluminum Alloy, Wrought.
QQ-N-288	- Nickel-Copper Alloy and Nickel-Copper-Silicon Alloy Castings.

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MIL-V-3	- Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of.
MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-R-17131	- Rods and Powders, Welding, Surfacing.
MIL-G-21032	- Gaskets, Metallic-Asbestos, Spiral Wound.
MIL-P-24503	- Packing Material, Graphitic, Corrugated Ribbon or Textured Tape and Preformed Rings.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-167-1	- Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-271	- Nondestructive Testing Requirements for Metals.
MIL-STD-278	- Fabrication Welding and Inspection; and Casting Inspection and Repair for Machinery, Piping and Pressure Vessels in Ships of the United States Navy.

2.1.2 Government drawing and publication. The following Government drawing and publication forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
803-1385620 - Handwheels for Valves.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
S9253-AD-MMO-010/VLVMAN - Valve Manual for 1200 and 600 PSI Steam Propulsion Systems.

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
B16.5 - Steel Pipe Flanges and Flanged Fittings. (DoD adopted)
B16.34 - Valves - Flanged and Buttwelding End. (DoD adopted)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
B 164 - Standard Specification for Nickel-Copper Alloy Rod, Bar, and Wire. (DoD adopted)
B 564 - Standard Specification for Nickel Alloy Forgings. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Materials. Materials shall be as specified in table I or shall be of equal or superior properties for the intended pressures and temperatures, and shall be selected so as to prevent galling, seizing or excessive wear on operating parts. Clearances shall be such as to prevent interferences due to thermal expansion. The valves shall be used for handling and continuous use with boiler water. Boiler water for the purpose of this specification is water having 500 parts per million (p/m) undissolved boiler sludge and 1150 p/m dissolved solids of which 500 p/m may be chlorides. The suitability of the material of all other parts not specified in this specification and the substitution of any of the

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specified material for a better material shall be subject to review by the Command or activity concerned and then its suitability shall be demonstrated by satisfactory performance of the valves during the tests specified in 4.3.

TABLE I. Valve assembly material.

Name of part	Material	Specification
Body bonnet	Nickel-copper	QQ-N-288, comp E
Valve disc	Nickel-copper	QQ-N-281
Valve stem	Nickel-copper-aluminum	QQ-N-286 - class A, age hardened (RC24-32)
Valve seat	Cobalt-chromium	MIL-R-17131
Stuffing boxes and glands	Nickel-copper	ASTM B 164 ASTM B 564 (UNS N04400)
Bolts, studs, cap screws	Nickel-copper-aluminum	QQ-N-286, class A
Nuts	Nickel-copper	QQ-N-281, class B
Gaskets	Spiral wound	MIL-G-21032, type I
Packing	Graphite	MIL-P-24503

3.2.1 Bodies, bonnets, and yoke retaining rings. Bodies, bonnets and yoke retaining rings shall be of nickel-copper alloy (monel) conforming to QQ-N-288, composition E.

3.2.2 Valve stems. Valve stems shall be of nickel-copper aluminum alloy in accordance with QQ-N-286, class A, age hardened (RC24-32).

3.2.3 Bolts, studs, cap screws, and nuts. Bolts, studs, and cap screws shall be nickel-copper-aluminum in accordance with QQ-N-286. Nuts shall be nickel-copper alloy in accordance with QQ-N-281, class B.

3.2.4 Handwheels. Handwheels shall be in accordance with Drawing 803-1385620. Handwheel material shall be aluminum alloy. Valves shall close by clockwise rotation of the handwheel when facing the handwheel.

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3.2.5 Hard seat (disc and seat). Valve seats and discs shall be faced by depositing a layer of cobalt-chromium alloy (stellite) conforming to type MIL-RCoCr-A-1 or MIL-RCoCr-A-2 of MIL-R-17131 with a finish thickness of hard faced surface of not less than 3/32 inch or shall be solid stellite discs and seats.

3.2.5.1 Valve discs. Valve discs shall be nickel-copper alloy in accordance with QQ-N-281, with seating area faced with cobalt-chromium composition (see 3.2.5) deposited by welding to ensure a positive bond between it and the disc, or shall be solid cobalt-chromium composition. The valve disc shall be of the swivel attachment type. Stellite shall be applied to the interface areas between the disc, stem and disc holder where disc rotation can cause galling. The disc shall incorporate a blow deflector feature which successfully completes the 200 cycle performance test (see 4.6.2.1) without seat damage (seat wear or wire drawing) and without seat leakage in excess of 15 cubic centimeters per hour (cm^3/h). A hardened washer (500 brinell minimum) or a pad of stellite shall be buttered to the inside of the disc at the point of contact, to prevent galling.

3.2.5.2 Valve seats. Valve seats shall be an integral part of the valve body or may consist of a tubular insert welded into the body with sufficient body material provided to permit the fitting of renewable seats, if necessary. Vacuum braze seats are unacceptable. The seating surface shall be faced with cobalt-chromium composition (see 3.2.5) or shall be solid stellite.

3.2.5.3 Stuffing boxes and glands. The packing gland shall conform to ASTM B 164 and ASTM B 564 (UNS N04400). The stuffing box shall be accessible without removal of stem or valve from the piping system. Adjustments of the packing gland shall ensure tightness under all operating conditions. Valve stem stuffing boxes shall be of ample depth to hold at least five turns of packing.

3.2.5.4 Packing and gaskets. Packing shall be graphite corrugated ribbon and preformed rings and conform to types I and II, class 1 of MIL-P-24503. Packing installation shall be in accordance with appendix B of NAVSEA S9253-AD-MMO-010/VLVMAN. Gaskets shall be spiral wound in accordance with MIL-G-21032.

3.2.6 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification, unless otherwise specifically specified.

3.3 General design. The valves shall be one piece integral body containing both the flanges and the valve body. Valves shall be designed to withstand nominal working pressures of 600 and 1500 lb/in^2 and to ensure tightness under severe service. Valves shall be designed in accordance with ANSI B16.34.

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3.3.1 Flange dimensions. The valve flange dimensions shall be in accordance with ANSI B16.5. For types I and III, inlet flange face to outlet flange face dimension shall be 9-1/2 inches. For types II and IV, the inlet flange face to the centerline and centerline to the outlet flange face dimension shall be 4-3/4 inches. For types V and VII, the inlet flange face to outlet flange face dimension shall be 12 inches. For types VI and VIII, the inlet flange face to centerline and centerline to outlet flange face dimension shall be 6 inches.

3.3.1.1 Flange finish. The gasket contact surfaces of flanges shall be finished with a circular lay (concentric or phonographic) having roughness not exceeding 500 roughness height rating (RHR) produced by machining cuts of 0.003 inch maximum depth but not less than 40 cuts per inch of face width.

3.3.1.2 Assembly dimensional tolerances. The flanged end connections shall be perpendicular to the centerline of the valve in both the X and Y axis. The maximum allowable tolerance is plus or minus 1/16 inch. The flange bolt holes shall be equally spaced and pairs of bolt holes shall straddle the vertical centerline of the valve. The maximum allowable flange rotation is plus or minus 1/16 inch. Flange dimensions shall be in accordance with ANSI B16.5, 600 and 1500 lb/in^2 as applicable.

3.3.2 Flow characteristics. Valves shall have an unrestricted area through the valves, with the valves in the full open position, and through all body passages. The minimum diameter of the body passages and seats shall be in accordance with annex A of ANSI B16.34. The pressure drop shall not exceed 30 feet equivalent length of pipe.

3.3.3 Valve bonnets. Valve bonnets shall be flanged and secured by bolts or studs and nuts or cap screws. The valve bonnet alignment shall be obtained by body guiding, that is, a close tolerance fit between bonnet and valve body. The manufacturer shall demonstrate ease of bonnet removal for valve repair. Bonnet flange face design shall assure proper compression of flange gasket when flanges are drawn up metal to metal.

3.3.4 Yokes. Valves shall be of the outside screw and yokes construction. Unless otherwise specified (see 6.2.1), yoke shall be integral with the bonnets. The yoke shall be fitted with a bushing, threaded to fit the stem. The yoke bushing material shall be manganese bronze, alloys B or E of QQ-C-523.

3.3.5 Stems. Valves shall have rotating stems with self-locking Acme threads. Stems shall turn clockwise to close the valve.

3.3.5.1 Stem angle. The angle between the centerline of the stem and the centerline through the outlet flange shall be within the range of 45 to 50 degrees.

3.3.6 Valve passage. The passage through the valves shall give a smooth flow through the valves without any abrupt changes in the direction of flow, in order to minimize the possibility of grease, scale and sludge from accumulating in the valves.

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3.3.7 Handwheel operation. Valves shall not exceed the allowable force as specified in table II to open and close while under their intended nominal working pressure. Maximum allowable operating forces for valve handwheels shall be as specified in table II. Valves for 600 lb/in^2 service (types I, II, III and IV) shall be fitted with 8 inch diameter handwheels; valves for 1500 lb/in^2 service (types V, VI, VII and VIII) shall have 12 inch diameter handwheels.

TABLE II. Maximum allowable operating forces for handwheels.

Handwheel diameter	Total tangential force on rim of handwheel	Equivalent torque
Inches	Pounds	Foot-pounds (Maximum tolerance plus or minus 5 foot-pounds)
8	124	41
12	180	90

3.3.8 Seat leakage. Allowable seat leakage is considered to be leakage of water not in excess of that specified when tested in accordance with 4.6.1.3, 4.6.1.4, 4.6.2.1 through 4.6.2.4.

3.3.9 Coldness. Valves shall pass the cold tests as specified in 4.6.1; operational tests (see 4.6.1.1), shell hydrostatic test (see 4.6.1.2) and seat leakage test (see 4.6.1.3).

3.3.10 Hydrostatic. Valves shall withstand the pressures specified in table III and shall not leak or seep externally when hydrostatically tested as specified in 4.6.1.2, 4.6.1.3 and 4.6.2.4.

TABLE III. Hydrostatic pressure.

ANSI rating	Shell pressure lb/in^2	Seat leakage pressure lb/in^2
600	1800	1320
1500	4500	3300

3.3.11 Welding and fabrication. Welding shall be in accordance with MIL-STD-278. Fabricated assemblies shall be stress relieved as units, prior to finish machining. The stress relieving shall be done in accordance with MIL-STD-278.

3.3.12 Shock and vibration.

3.3.12.1 Shock. The valves shall not be damaged or malfunction when tested as specified in 4.6.3.1.

3.3.12.2 Vibration. Valves shall not be damaged or malfunction by externally imposed vibrations up to 50 hertz (Hz). Valves shall withstand the vibration test specified in 4.6.3.2.

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3.3.13 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.4 Body markings. Valve bodies shall have the pressure rating and the manufacturer's name or trademark cast or forged integral with the valve body. Globe valves, types I and V, shall have a bridge wall marking in addition to the above markings. Stop check valves, types III and VII, shall have a flow arrow in addition to the above markings.

3.4.1 Identification plate. Valves shall bear an identification plate made of corrosion resistant steel, anodized aluminum or brass. Identification plate shall include the following information:

(a) Manufacturer's name or trademark.
(b) Valve size and rating.
(c) Body, trim and seat material.
(d) Manufacturers' identification number (Federal Supply Code or Manufacturer's FSCN).
(e) DOD-V-24675 and type.
(f) Component identification number (CID).
(g) National stock number (NSN).
(h) When applicable, a "no backseat" warning.
(i) Date of manufacture.

3.5 Drawings. In addition to the drawing content required by the data ordering document (see 6.2.2), the following unique technical features shall be included:

(a) Accurately scaled sectional assembly which clearly depicts the design and construction of the valve.
(b) Bill of material listing specification, grade, condition, and any other data required to fully identify the properties of the materials.
(c) Dimensions - overall, accessibility space, including disassembly clearances, and all other dimensions pertinent to installation.
(d) Estimated weight and center of gravity and any limitations on installation of valve which must be noted and approved by the contracting activity prior to manufacture of the first valve.
(e) Reference to any previous shock and vibration for valve and test report numbers.
(f) Details of the seat, disc, and stem assembly and all other replaceable internal trim.
(g) Recommended assembly torques, or equivalent procedures, for making up all joints and threaded assemblies.
(h) Table of seal dimensions and a tabulation of required gasket characteristics including all dimensions (with tolerances) and load versus compression characteristics (with tolerances).

3.6 Special tools. Valves shall be designed so that special tools are not required for installation or maintenance. Special tools are defined as those tools not listed in the Federal Supply Catalog. (Copies of this catalog may be consulted in the office of the Defense Contract Administration Services Management Area (DCASMA)).

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3.7 Workmanship. Valves shall be free from defects affecting their appearance or operation. Castings shall be clean, sound and free from blow holes, head spots, porosity, cracks and other injurious defects. They shall be smoothed and well cleaned both inside and outside. All fins and roughness shall be removed. Castings shall not be repaired, plugged, impregnated, brazed or burned, unless otherwise specified in MIL-STD-278.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination specified in 4.5 and the tests specified in 4.6.1 through 4.6.3.2.

4.3.1 Test report. The contractor shall prepare a first article test report in accordance with the data ordering document included in the contract or order (see 6.2.2).

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination in 4.5 and the tests specified in 4.6.1 through 4.6.1.6.

4.4.1 Lot. Valves of the same size, style and type (see 1.2) offered for delivery at the same time and manufactured within the preceding 12 months shall be considered a lot for purposes of sampling.

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4.4.2 Sampling for visual and dimensional examination. Sampling of each lot for visual and dimensional examination specified in 4.5 shall be conducted in accordance with MIL-STD-105, general inspection level II, acceptable quality level of 2.5 percent defective.

4.4.3 Sampling for test. Each valve in the lot shall be tested as specified in 4.6.1.1 through 4.6.1.6.

4.5 Examination. Each of the sample valves selected in accordance with 4.4.2 shall be visually and dimensionally examined to determine conformance with the requirements of the specification as to material, workmanship, and dimensions specified on the drawings, and for such modifications as may be required as a result of first article inspection (see 4.3). Material examination shall be as specified in 4.5.1 and 4.5.1.1.

4.5.1 Pressure containing parts shall be examined to determine conformance with the applicable material specification.

4.5.1.1 Certification that parts conform to their applicable material specification may be accepted providing that such certification contains the actual examination, test, or other verifiable data (see 4.7.1). Certification representing stock more than 18 months old is not acceptable (see 6.2.2).

4.6 Tests.

4.6.1 Cold tests.

4.6.1.1 Operational test. Valves shall be cycled 10 times unpressurized. Maximum breaking and running torques shall not exceed the values specified in table IV.

TABLE IV. Maximum breaking and running torque.

Handwheel diameter inches	Breaking torque	Running torque
8	41	20
12	90	45

4.6.1.2 Shell hydrostatic test. Each valve shall be hydrostatically tested for body strength and porosity. The test fluid shall be clean fresh water at a temperature between 40 and 100 degrees Fahrenheit (°F). The test pressure shall be as specified in table III. The test duration shall be 10 minutes minimum. The valves shall be in the open position during the test. No external leakage or seepage is permitted and any permanent deformation shall be cause for rejection.

4.6.1.3 Seat leakage test. Valves shall be pressurized in the direction tending to open the valve and examined for seat tightness. The valves shall be torqued closed prior to the test. Torque values shall not exceed those specified in table II. The test pressure shall be as specified in table III. The duration of the test shall be 3 minutes minimum. There shall be no visible signs of leakage. Should any visible signs of leakage be seen, the test shall be continued until an accurate leakage rate can be determined. Leakage shall not exceed 15 cm^3/h.

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4.6.1.4 Reverse flow seat leakage test. In addition to the tests specified in 4.6.1.1 through 4.6.1.3, stop check valves, types III, IV, VII and VIII shall be hydrostatically tested in the reverse direction to verify the check feature. The test shall be conducted with the valve in the open position. The test pressure shall be 50 and 1200 lb/in^2 for 1500 lb/in^2 service valves and 50 and 600 lb/in^2 for 600 lb/in^2 service valves. The test duration shall be 3 minutes minimum. Should any visual leakage occur, the test shall be continued until an accurate leakage rate can be determined. The maximum allowable leakage is 37 cm^3/h.

4.6.1.5 Stellite seating. Stellite seating surface shall be liquid penetrant inspected in accordance with MIL-STD-271 after rough machining and shall be free of cracks and crack-like defects.

4.6.1.6 Equivalent torque. Tests for compliance with table II shall be made using 100 percent hydrostatic test condition.

4.6.2 Performance test.

4.6.2.1 200 cycle test. The 200 cycle test shall be conducted with boiler water at the pressure and temperature of the intended service, blowing down to atmospheric pressure. The blow down cycle shall consist of opening the valve under test, rapidly cycling the upstream boiler blow valve and then closing the test valve. The time between cycles shall be 5 minutes minimum. After each 20 cycles, seat leakage tests shall be conducted. The test valve shall be closed and torqued to the value specified in table II prior to the seat leakage tests. The maximum allowable leakage is 15 cm^3/h. At 50 cycle intervals, maximum handwheel break and running torques shall be measured. The torque values shall not exceed those specified in table II.

4.6.2.2 Warm-up trials. Upon completion of the 200 cycle test, four warm-up trials shall be made to determine that the valve remains tight, after the valve has been closed, during and after cooling to room temperature (65 to 95°F) and during warm-up to saturated temperature conditions. The operating torque shall be taken periodically to assure that a man can open and close the valve whether the valve was closed at room temperature or at the operating temperature of the boiler. The four warm-up trials shall be in accordance with the following schedule:

(a) Two tests at a boiler warm-up rate of 2 hours.
(b) Two tests at a boiler warm-up rate of 30 minutes.
(c) Valve seat leakage shall not exceed 15 cm^3/h during warm-up, following cold closure when the above tests are run and the closure torque is as specified in table II.

4.6.2.3 Reverse flow test. After completion of the warm-up trials, stop check valves, types III, IV, VII and VIII shall be reversed in the test loop and pressurized to boiler operating conditions. A seat leakage test shall be conducted. The maximum allowable seat leakage shall be 37 cm^3/h. The duration of the test shall be 30 minutes.

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4.6.2.4 Emergency operational test. The valves shall be cycled 5 times against boiler pressure. Break and running torque values shall be measured. Torque values shall not exceed those specified in table IV. After completion of the 5 cycles, a seat leakage test shall be conducted. The test duration shall be 30 minutes. The leakage rate shall not exceed 15 cm^3/h.

4.6.2.5 Final inspection. After completion of tests specified in 4.6.1 through 4.6.2.4, the valve shall be disassembled and inspected for indications of galling, wear and mechanical defects.

4.6.3 Shock and vibration.

4.6.3.1 Shock. One valve of each type and new design shall be tested in accordance with grade B, class 1, type A of MIL-S-901.

4.6.3.2 Vibration. One valve of each type and new design shall be subjected to the vibration tests in accordance with type I of MIL-STD-167-1.

4.7 Verification of material identification and control. Valves will be used in critical shipboard systems. The use of incorrect or defective material would create a high probability of failure resulting in serious personnel injury, loss of life, loss of vital shipboard systems, or loss of the ship itself. Therefore, valves shall require material and special control procedures that are invoked to ensure receipt of correct material.

4.7.1 Materials which form the pressure boundary valves shall be controlled by the requirements specified in 4.7.1.1.

4.7.1.1 Quantitative chemical and physical analysis of material traceable to heat identification shall be required for the following. Certification documentation shall include, in addition to the material specification: class, form, condition, grade, type and composition, as applicable to material supplied.

(a) Valve body.
(b) Valve bonnet.
(c) The valve stem which penetrates the pressure boundary.
(d) Metallic pressure boundary internal parts.
(e) Threaded fasteners, 1/2 inch nominal and larger.
(f) Pressure seals, such as O-rings are specifically excluded for material control requirements.
(g) Questions regarding applicability of material requirements shall be referred to NAVSEA.
(h) A statement of hydrostatic tests performed giving test pressures.

4.8 Inspection of packaging. Sample packages and packs and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified herein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.5.)

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5.1 Preservation, packing and marking. Valves shall be individually preserved-packaged level A, C or commercial, packed level A, B, C or commercial and marked in accordance with MIL-V-3, as specified (see 6.2.1).

5.2 Data (see 3.5 and 6.2.2). Drawings shall be prepared for shipment in accordance with the applicable specifications. Other data, reports, plans and so forth, shall be prepared for shipment in accordance with the level C requirements of MIL-V-3.

6. NOTES

6.1 Intended use. The valves covered by this specification are intended for use in shipboard boiler blowdown systems other than the first valves off the boiler draw and headers.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of valves (see 1.2).
(c) If first article inspection is required (see 3.1).
(d) Yoke construction (see 3.3.4).
(e) Selection of applicable levels of preservation and packing required (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.5	Drawings, engineering and associated lists	DI-E-7031	Level 3 Design activity designation - Government Drawing number - Government
4.3.1	First article inspection report	DI-T-4902	----
4.5.1.1	Certificate of compliance	DI-E-2121	----

(Data item descriptions related to this specification and identified in section 6, will be approved and listed as such in DoD 5010.12-L., Vol. I, AMSDL. Copies of the data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article. When a first article inspection is required, the item should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Provisioning. Provisioning Technical Documentation (PTD), spare parts and repair parts should be furnished as specified in the contract.

6.4.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.5 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Subject term (key word) listing.

Angle design
Boiler blowdown systems
Flanged angle
Flanged globe
Stop check
Straightway design
Monel valves

Preparing activity:
Navy - SH
(Project 4820-N499)

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